

Supplementary Material: Patient-Visit-Spanned Hypergraph Learning for EHR-based Diagnosis Prediction

A Code Link

The code is available at '<https://github.com/Forwhyy/PVHGL>'.

B Details of Baselines

The following baselines are considered for comparison, which are categorized into three groups based on how they encode clinical information: (i) sequence-based models that capture temporal dynamics across clinical visits, (ii) GNN-based models that emphasize pairwise relations between medical codes, and (iii) HGNN-based models that are designed to capture higher-order interactions among medical codes. This diverse set of baselines enables a comprehensive evaluation of our method.

Sequence-based models

- **RETAIN** introduces a RNN with a reverse-time attention mechanism, designed to process clinical visit sequences while assigning interpretable weights to both visits and medical events.
- **Transformer** is a neural network architecture that relies on self-attention mechanisms and positional encoding to model sequential data.
- **StageNet** introduces an LSTM-based framework that segments visit sequences into stages and applies dual attention mechanisms to capture both intra- and inter-stage temporal dependencies in irregular clinical time series.
- **HiTANet** first employs a time-aware Transformer to inject time intervals into visit representations and derive visit-level (local) attention. It then computes global attention over visits, and fuses the two attentions to form the patient representation for prediction.

GNN-based models

- **CGL** constructs a patient-disease observation graph to extract structural features from EHR, along with a code ontology graph to model horizontal relationships between medical codes.
- **SafeDrug** constructs a molecular graph for each drug and a drug-drug interaction graph to model adverse combinations, and employs three separate RNNs to capture dynamic patient history.

- **Chet** develops a global disease co-occurrence graph and three dynamic subgraphs based on individual clinical visits, seamlessly integrating global patterns with local dynamic information.
- **Sherbet** designs disease complications graph and a multi-level attention method to compute the contributions of diseases and admissions.
- **ADRL** builds a disease-disease relation graph to learn adaptive disease embeddings, and couples them with a visit-sequence encoder to model patient history for diagnosis prediction.

HGNN-based models

- **MCHN** constructs three types of hypergraphs and conducts multi-channel hypergraph message passing, where code- and visit-level attention fuse cross-view representations for diagnosis prediction.
- **HypEHR** models EHR as a hypergraph and employs self-attention as the message passing function within a two-step message propagation framework.
- **AllSet** unifies hypergraph learning by parameterizing both node-to-hyperedge and hyperedge-to-node message passing as multiset functions. With alternating set aggregations, it iteratively refines node embeddings.
- **TACCO** proposes a task-guided hypergraph co-clustering framework that jointly clusters clinical concepts and clinical visits to improve prediction tasks.
- **Shy** models each patient's longitudinal EHR as a patient-level hypergraph and applies hypergraph message passing to capture high-order code interactions for diagnosis prediction.

The key differences among these baselines in terms of construction, spatial propagation, and whether they explicitly leverage a medical code co-occurrence matrix for guided learning are summarized in Table 1.

C Experimental Setup

C.1 Baselines Configuration

- All experiments are performed on a single GPU-enabled machine with an NVIDIA GeForce RTX 3050.

Model	Construction	Spatial propagation	Medical code co-occurrence matrix-guided learning
RETAIN	Visit sequence	×	×
Transformer	Visit sequence	×	×
StageNet	Visit sequence	×	×
HiTANet	Visit sequence	×	×
CGL	Patient-disease graph and diseases graph	Adjacency-based GNN propagation	×
SafeDrug	Drug molecular graph	Adjacency-based GNN propagation	×
Chet	Diseases graph and diagnosis subgraphs	Adjacency-based GNN propagation	×
Sherbet	Diseases graph	Adjacency-based GNN propagation	×
ADRL	Diseases graph	Adjacency-based GNN propagation	×
AllSet	Cohort-level hypergraph	Localized two-step propagation	×
HypEHR	Cohort-level hypergraph	Localized two-step propagation	×
MCHN	Patient-specific hypergraph	Localized two-step propagation	×
TACCO	Cohort-level hypergraph	Localized two-step propagation	×
Shy	Patient-specific hypergraph	Localized two-step propagation	×
PVHGL	Cohort-level hypergraph	One-step propagation enhanced by two structure encoding matrices	✓

Table 1: Detailed information of baselines.

- All compared models share the same termination criterion, where early stopping is applied if the performance on the validation metric does not improve for 10 consecutive epochs.
- The optimizer for each baseline follows the configuration provided in its corresponding original paper.
- All experimental results reported in this paper are averaged over five runs with different random seed initializations.

C.2 Hyperparameter Setting

All the compared models are carefully tuned by a grid search to obtain the best performance with learning rate $\eta \in \{1e^{-5}, 1e^{-4}, 1e^{-3}, 2e^{-3}, 3e^{-3}, 4e^{-3}, 5e^{-3}, 1e^{-2}\}$, regularization coefficient $\lambda \in \{1e^{-3}, 1e^{-2}, 1e^{-1}, 2e^{-1}, 3e^{-1}, 4e^{-1}, 5e^{-1}\}$, the embedding dimension $d \in \{32, 64, 128, 256, 512\}$, the layers $l \in \{1, 2, 3, 4, 5\}$. For the proposed PVHGL, other hyperparameter like control coefficient is also tuned via grid search with $\alpha \in \{0.01, 0.1, 0.2, 0.3, 0.4, 0.5\}$.

D Parameter Sensitivity Analysis

D.1 Effects of embedding

As illustrated on the left portion of Figure 1, the performance of PVHGL improves with increasing embedding dimension at the beginning, which indicates that large embedding dimension contributes to a strong representational capacity and facilitates more effective feature learning. However, when the embedding dimension exceeds a certain threshold, the performance starts to decline. This is mainly because an excessively large embedding dimension may lead to overfitting or introduce irrelevant or redundant information, thus hindering the model’s ability to learn discriminative representations.

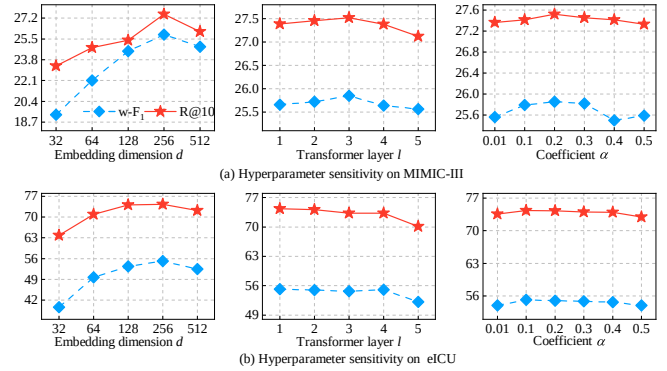


Figure 1: Hyperparameter analysis on the MIMIC-III and eICU datasets.

D.2 Effects of Transformer layers

As depicted on the middle portion of Figure 1, we observe that PVHGL achieves the best overall performance when the number of Transformer layers is set to 3 on MIMIC-III. This is mainly because a lower number of layers is insufficient to effectively capture complex dependencies among clinical events. As the number of layers further increases, the performance begins to degrade. This occurs primarily because the attention distributions across different layers gradually converge, eventually leading to attention collapse.

D.3 Effects of control coefficient

Considering the control coefficient α , it governs the influence of the medical code co-occurrence matrix. As shown in the right panel of Figure 1, it is clear that moderate values (e.g., 0.2–0.3) yield the most favorable performance. This suggests that integrating co-occurrence patterns offers clinically meaningful guidance to the learning process. However, when the

Code	System	Clinical description
37.31	ICD-9 (Proc.)	Pericardiectomy
88.72	ICD-9 (Proc.)	Diagnostic ultrasound of heart
39.61	ICD-9 (Proc.)	Extracorporeal circulation auxiliary to open heart surgery
458.9	ICD-9 (Dx)	Hypotension, unspecified
785.51	ICD-9 (Dx)	Cardiogenic shock
511.9	ICD-9 (Dx)	Pleural effusion, unspecified
423.9	ICD-9 (Dx)	Disease of pericardium, unspecified
A12AA	ATC (L4)	Calcium
A06AA	ATC (L4)	Softeners, emollients (laxatives)
A02BA	ATC (L4)	H ₂ -receptor antagonists

Table 2: Code-to-description mapping for the ICD-9 and ATC (level-4) codes used in our attention heatmap visualization. ATC codes are reported at the chemical subgroup (level-4) rather than individual active ingredients. Abbreviations: Dx = diagnosis; Proc. = procedure; L4 = ATC level-4.

value of α is too small, PVHGL fails to sufficiently leverage the relational prior. Conversely, an excessively large α may cause the representations to be dominated by co-occurrence statistics, thereby overlooking other informative interactions among medical codes.

E Code-to-Description Mapping for Visualization Labels

For better clinical interpretability, we convert the ICD-9 and ATC codes used in our qualitative visualizations (e.g., attention heatmaps) into their corresponding human-readable clinical descriptions, as summarized in Table 2. For ICD-9 diagnosis/procedure codes, we follow the official ICD-9 codebook; for medications, we use the ATC index. Note that ATC codes are reported at level-4 (chemical subgroup), thus each ATC label corresponds to a medication category rather than a specific active ingredient.